

Process book

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1 Project goal

The goal of this project was to explore representation patterns in the book industry through data, visualizing the data in different and creative ways to obtain distinct insights. Many aspects are touched upon in our visualization but we placed an important emphasis on gender differences and disparity in the book industry. This focus was translated into creating multiple visualizations comparing female and male authors in different ways.

2 Datasets and data processing

This project was built with two main datasets, [International bestsellers](#) and [Goodreads](#). These datasets differ in their content, as International Bestsellers has data about authors' nationality and gender as well as publisher's names. The Goodreads dataset has the pricing and rating information as well as the genres of books.

One thing we tried to do after our initial data analysis, was to add genders to the Goodreads dataset. This was done using a python package named gender guesser as well as using the authors from the International Bestsellers dataset. This allowed us to successfully map the gender of 50632 authors out of the 63700 available in the Goodreads dataset. We have chosen not to include the authors that were mapped to androgynous, mostly_male and mostly_female, as it was hard to sort which were accurate and which where not. It was also too time consuming to do by hand. We tried seeing if there where any implementations of a gender guesser that used machine learning but most where outdated and performed poorly.

Given more time we believe we could have found a better way to infer the gender of the authors, as we think that the python package might have performed better on traditional english names, thus making us remove authors with names from other origins.

Several others aggregated datasets were created from this basis to suit the needs of our individual visualizations.

3 Visualizations created

3.1 Filtering Gender

The goal of this visualization was to let users directly compare the number of books attributed to male, female, and other authorship groups. This gives users a clear visual image of the disparity in the gender of authors for certain book genres or nationalities.

The visualization is built around a simple yet elegant bar chart. Users can choose whether they want to filter by genres or by author nationalities, then select the categories they want to include. When several categories are selected, the visualization does not display one bar chart per category. Instead, it combines all selected categories into a single total and updates the three bars accordingly.

We included a minimum-books slider to also give the user more flexibility. Some nationalities appear very few times in the data, which can make the interface a bit crowded, but it can also be important to get some insight from those groups that can be underrepresented.

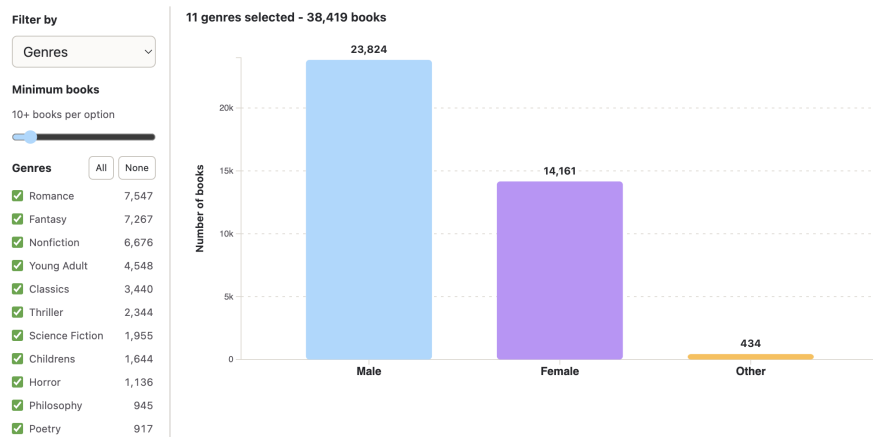


Figure 1: The finalized visualization of Filtering Gender

3.2 The Gender Scatter

The goal of this visualization was to compare whether books associated with male and female authorship groups differ in terms of average rating and average price. We wanted to look at this question across two kinds of categories: book genres and author nationalities. Each point in the scatterplot represents the weighted average for the currently selected categories, with the x-axis showing average rating and the y-axis showing average price.

The visualization was designed to make comparison easy rather than to show every category at once. Users can switch between grouping by genre and grouping by nationality, then select or remove categories from the filter list. This lets them see both the overall pattern and more specific subsets. For example, selecting all genres gives a broad view of price and rating differences between the two gender groups, while selecting only a few genres makes it possible to inspect whether the difference is driven by particular literary categories.

A large part of the design process was simplifying the interaction. In an earlier version, we had a slider controlling the minimum number of books required in each gender group. The slider ranged from 2 to 50 books, with a default of 5. Although this gave more control, it also made the visualization feel more complicated than necessary. It was especially confusing for nationalities, where many categories have very few books and changing the slider could make many different options appear, completely bloating the visualization. We eventually removed the slider and replaced it with a fixed threshold, which made the interface cleaner and easier to understand.

On top of this, the scatterplot used to show two separate points, one for male authors and one for non-male authors, for every individual category. This meant that selecting many genres or nationalities quickly produced a large number of points on the chart. While this was more detailed, it also made the visualization hard to read, especially when several categories overlapped or appeared close together. We therefore changed the design so that the selected categories are aggregated into one weighted average point per gender group. This sacrificed some category-level detail, but made the comparison much clearer and prevented the chart from becoming visually overloaded.

The final version keeps only categories where both gender groups have at least two books. This limit is important because categories with only one book would make the average rating or price too dependent on a single title. The threshold is deliberately low, because nationality data is much sparser than genre data. With a higher threshold, most nationalities would disappear, making that view much less useful. After applying this rule, all 11 selected genres remain available, while the nationality view is reduced from 54 possible nationalities to 11 comparable ones.

Another limitation is that ratings are very compressed. Most books are rated between 3 and 5, so even small horizontal differences can look meaningful but should be interpreted carefully.

The scatterplot is best understood as an exploratory view: it helps reveal possible differences in gendered pricing and rating patterns, but it does not prove a causal relationship.

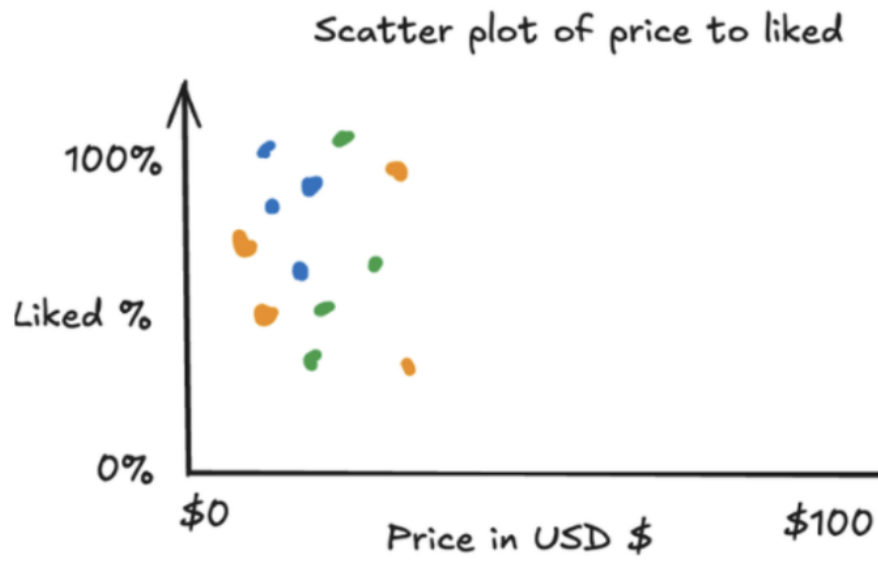


Figure 2: The initial sketch for the Gender Scatter plot

3.3 Exploring gender

This visualization focuses on comparing the gender distribution of authors across genres and nationalities. The ability to sort is an important feature we wished to highlight with this visualization because it can really change the perspective on the data and make things clearer to understand in terms of the difference between male and female authors.

We also had to make design decisions to keep the chart readable. Initially, we experimented with a minimum-books slider, present in some of the other visualizations. But this became problematic for the nationalities. When the threshold was too low, the chart included many small categories, and the visualization became very crowded.

The visualization still includes a row slider, but it's purpose is to control the number of rows that are displayed. This allows the user to select a small number of rows for clarity, while modifying the sorting feature to extract broad insights. But also they can choose a larger number of rows to have the full picture.

3.4 Authors Across The World

The goal of this visualization was to explore which author's nationalities are most prominent in the world. The idea was to scale the countries in the map by their number of authors. A sketch of our initial idea can be seen in 3a. Luckily for us, there were two helpful implementations on GitHub, [d3-cartogram](#) and [topogram](#).

We had to adapt a few things to be able to make it work on our Hugo website. First, we had to find a dataset to be able to connect our [International bestsellers](#) dataset with the [TopoJSON](#) data to represent the map of countries. One challenge we ran into when creating this cartogram was anti-meridian countries, like Fiji and Russia. These countries when drawn would create lines across the map, as shown in 3b. To solve they had to be drawn and scaled separately in the code, as to not interact with the other countries. Another challenge was mapping countries to their country code. This was solved by using a dataset with the different possible country codes and their name, from another [GitHub repository](#). This leads us to another challenge which is the multiple ways to write the name of a country, for example, Turkey and Türkiye. To be able to map them to their country code and then to their TopoJSON data, we had to handle them separately.

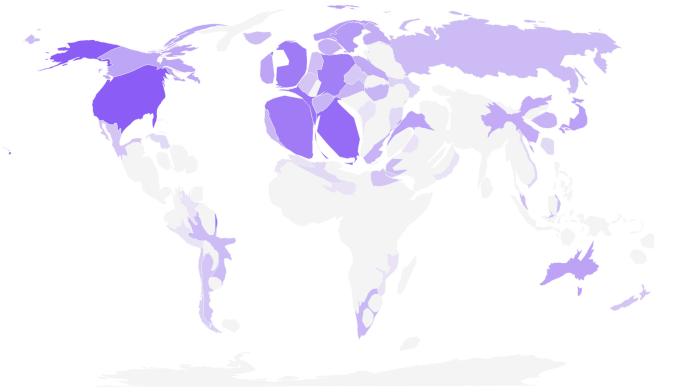
scale countries by amount of books published by author's from there



(a) The initial scaled map idea



(b) Illustration of the anti-meridian issue



(c) Final cartogram

3.5 The Countries Race

The idea behind this bar graph was to be able to better understand which countries have the most authors per million of inhabitants. This would allow us to put into perspective the scaling seen in the Authors across the world map. This chart was first a simple bar chart, that was later turned into a race. We have chosen to only represent the first 10 countries in the race as otherwise the visualization became too noisy. We reused the data from this [GitHub repository](#), to be able to map countries to their different codes and be able to represent them as their corresponding emoji flag.

To be able to scale by population we used data from [World Development Indicators](#). After testing the countries race a few times we felt it would benefit from allowing the user to toggle between the scaled and unscaled versions to better understand how authors are distributed across populations. We then added a speed bar to be more adaptive to users needs.

3.6 The Bestseller's Timeline

The goal of this visualization was to show how highly rated bestseller books are distributed over time, while comparing women-led and men-led titles in a direct and readable way. To make this comparison clear, we placed women-led titles above the central timeline and men-led titles below it. Each year shows the representative top-rated title for each group, together with its author and rating. By hovering or focusing on a year, the user can see more detailed information, including the number of rated titles, the number of bestseller chart appearances, and the best-rated title for that year. This makes the visualization both an overview of long-term trends and a way to inspect specific moments in the data.

A main design decision was to use publication year rather than bestseller ranking year. The bestseller dataset covers monthly international rankings from 2013 to 2022, but many books in those rankings were originally published much earlier. Using publication year allowed us to build a cultural timeline of when these successful books first appeared, instead of only showing when they became commercially visible in the ranking data.

We also had to define some data limits. The bestseller data was merged with the Goodreads-style book dataset in order to obtain ratings and publication dates. Books without a valid rating or a parsable publication year were removed, since they could not be positioned or compared meaningfully. Another important limitation was sparsity. Some years have only men-led titles, some have only women-led titles, and some years have no rated books after filtering. Instead of interpolating or filling these gaps, we kept the missing years visible as absences.

3.7 The Bookshop

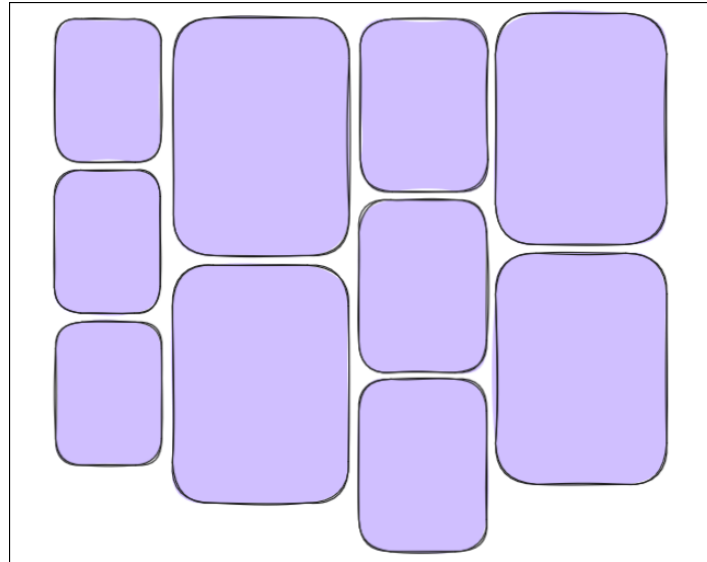
The goal of this visualization is to help users discover books from Goodreads written by women across a variety of languages, a filtering option rarely offered by conventional book discovery platforms. The initial concept took the form of a scrollable gallery (figure 4a), but this approach proved insufficiently engaging: endlessly scrolling through cover images offers little interactivity. We therefore pivoted toward a more immersive format, ultimately settling on a virtual bookshelf.

Users can refine their search using filters for language, genre, and rating. Once applied, up to 15 books matching the selected criteria are sampled and placed onto the shelves via a smooth animation. Hovering over a book reveals its rating and genres at a glance, while clicking opens its synopsis for a deeper look.

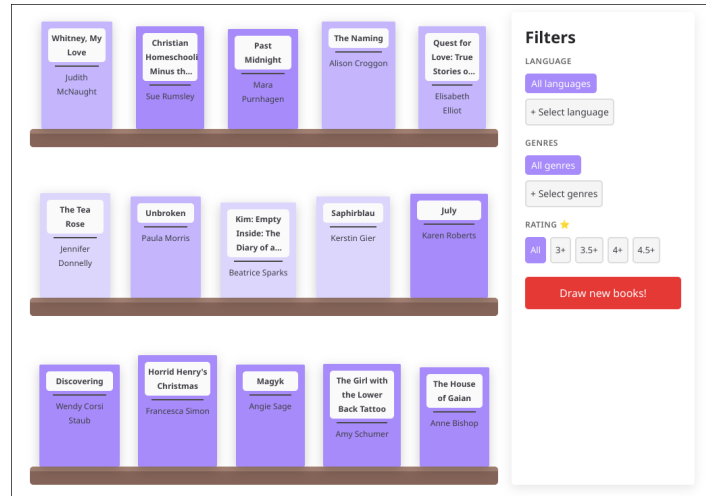
Crucially, the random sampling keeps the experience fresh: each new query yields a different selection, encouraging users to explore and return. The final result is shown in figure 4b.

3.8 The Most Published Languages

This visualization displays the proportion of books written in each language as a waffle chart, a format well-suited to conveying differences in scale. A legend accompanies the chart with exact percentages per language. The chart originally represented the proportion of prizes received by gender, but we found this comparison problematic: literary prizes are often genre-specific and restricted to residents of certain countries, introducing hard-to-detect biases. Language distribution proved to be a more meaningful and reliable metric. The result (figure 5) yields a striking insight: Goodreads books



(a) The initial book discovery idea



(b) The bookshop final result

Figure 4: Book discovery interface

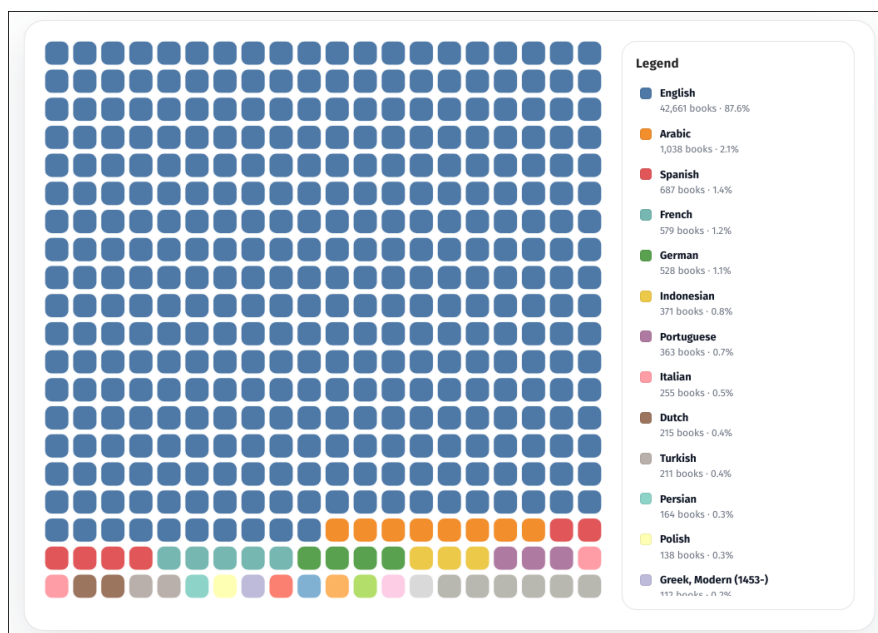


Figure 5: The language distribution of published books in goodreads

are overwhelmingly in English, with Arabic as the second most represented language — a surprising finding, as we had expected Spanish to rank higher.

3.9 Equality By Publishers

In this visualization, we wanted to uncover gender bias in books published by prominent publishers. For this purpose, a diverging chart from 50% seemed to be most appropriate, as it shows in a glance how far from gender equality a publisher resides.

The initial sketch is shown on figure

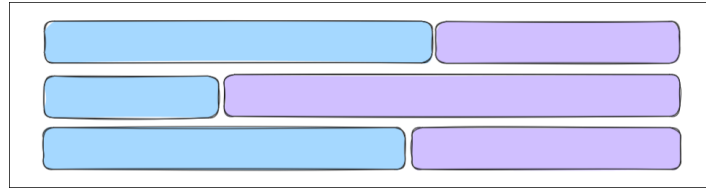
Looking at the chart, we were surprised at how far from the middle certain results were, so we looked for a possible explanation. By researching the publishers with the most extreme results online, we realized that the publishers with the highest published women ratio were mostly focused on romance and young adult books, whereas publishers with mostly men books were editing science fiction and fantasy. To explore this aspect further, we added for each publisher the three most represented genres, and this seems to confirm the hypothesis of genres having a big influence on gender equality.

Finally, we thought it could be insightful to see what the situation is for the biggest publishers, so we added an option to sort by number of published books, which shows that the largest publishers are still mostly male-dominated.

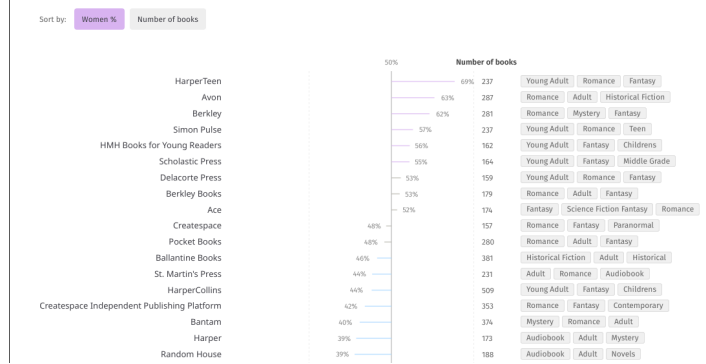
4 Results

Our visualizations revealed several consistent patterns in the publishing industry. The most striking finding is the strong link between genre and author gender: romance and young adult fiction are disproportionately written by women, while science fiction and fantasy remain heavily male-dominated. This was visible across multiple views, from the *Filtering Gender* bar chart to the *Equality By Publishers* diverging chart, where the most skewed publishers turned out to specialize in precisely these genres. The largest publishers, despite their broader catalogs, still lean male overall.

Geographic disparities were also notable. Countries like Japan show a far wider gender gap than others, suggesting that the imbalance is not uniform and that cultural and regional factors play a meaningful role beyond what aggregate global statistics reveal. The *Authors Across the World* visualizations showed that the proportion of authors published by the US and bigger European countries is far greater than those from Asian or African countries. The *The Countries Race* put this result into



(a) Initial sketch of the publisher ratio visualization



(b) Final implementation

Figure 6: Publisher and gender visualization

perspective by putting some smaller countries with a higher number of authors in the forefront, like Iceland and Switzerland.

Finally, our data has a significant structural limitation worth highlighting: the Goodreads dataset is overwhelmingly English-language and oriented toward North American and European readers. This means our results likely reflect the publishing landscape of those regions more than a truly global picture, and findings related to nationality or language should be interpreted with that in mind.

5 Conclusion

This project set out to explore representation patterns in the book industry, and the visualizations we built collectively paint a nuanced picture of an industry still marked by significant imbalances.

A few themes emerged consistently. Genre appears to be one of the strongest predictors of author gender, with romance and young adult fiction skewing toward women while science fiction and fantasy remain male-dominated, a pattern visible across both the publisher equality chart and the filtering gender visualization. Geography also plays a role: certain countries, notably Japan, show far greater gender gaps than others, suggesting that cultural factors shape representation in ways that global aggregates can obscure.

Our findings come with important caveats. The datasets are heavily biased toward English-language books and readers based in North America and Europe, limiting what we can say about publishing traditions elsewhere. Gender assignment through automated tools is also imperfect, and our decision to exclude ambiguous cases means some authors are absent from the analysis entirely.

Several visualizations, particularly the gender scatter and the bestseller timeline, are best understood as exploratory tools that surface patterns worth investigating, rather than definitive evidence of causal relationships.

We hope these visualizations offer a starting point for conversations about equity in publishing, and demonstrate that even imperfect data, explored thoughtfully, can reveal structures that are otherwise easy to overlook.